

Q1: How predictable is a player's next game?

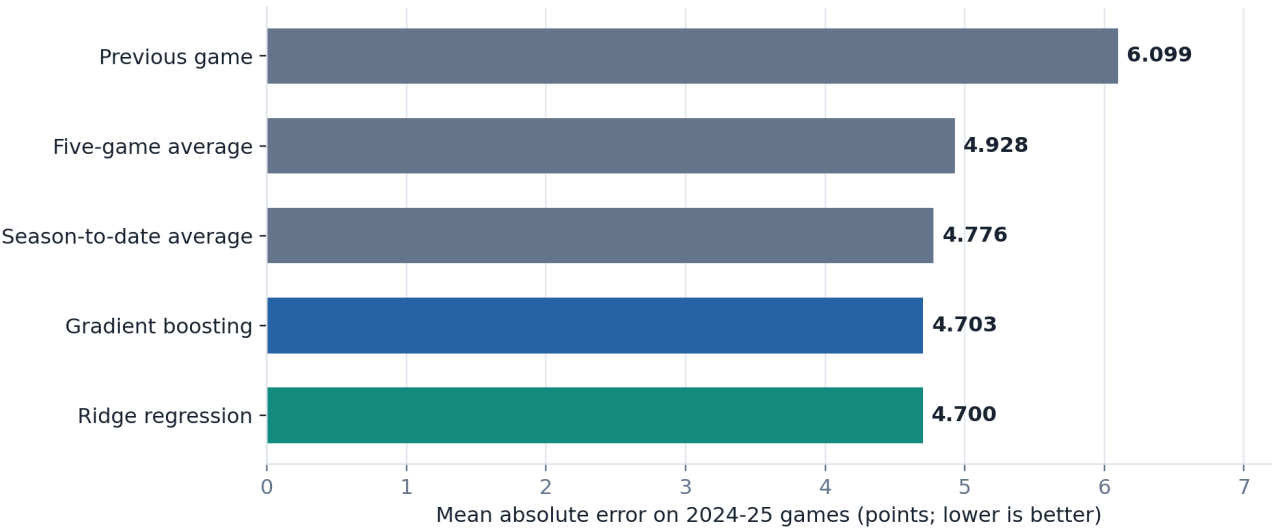
We compare the models with a simple player baseline and test them on a later season.

Setup

We trained on 2019-20 through 2023-24 and froze the models before evaluating **20,914 player-games** in 2024-25. Every rolling feature is shifted, so it only uses information from earlier games. We compare the models with each player's season-to-date scoring average, which is a stronger baseline than the league average. We only predict games after the player has made 10 appearances in that season.

The models only slightly beat a player's season average

Tested on 2024-25; the best model improves MAE by 0.08 points (95% CI: 0.05 to 0.10)



Source: NBA Stats API. Test: 20,914 games from 502 players in 2024-25.

Figure 1. Mean absolute error on the 2024-25 season. Lower is better. The two machine-learning models have almost the same error and are only slightly better than the player baseline.

Main result: Ridge regression reaches 4.70 MAE, which is 0.08 points better than the season-average baseline (95% CI: 0.05 to 0.10). The confidence interval does not include zero, but the difference is too small to call the predictions precise.

Q1: Who is hardest to predict?

Absolute error and relative error tell different stories about stars and role players.

Prediction error changes by scoring role



Figure 2. Scoring roles are assigned from pre-game season-to-date points. Gray is the running-average baseline; green is Ridge regression. Sample sizes are printed under each group.

Stars have the largest error in raw points (6.47), but the smallest error relative to their scoring level (27%). Role players reverse that pattern: only 3.70 points of absolute error, yet 64% relative error. This means the answer depends on whether error is measured in points or as a percentage of scoring.

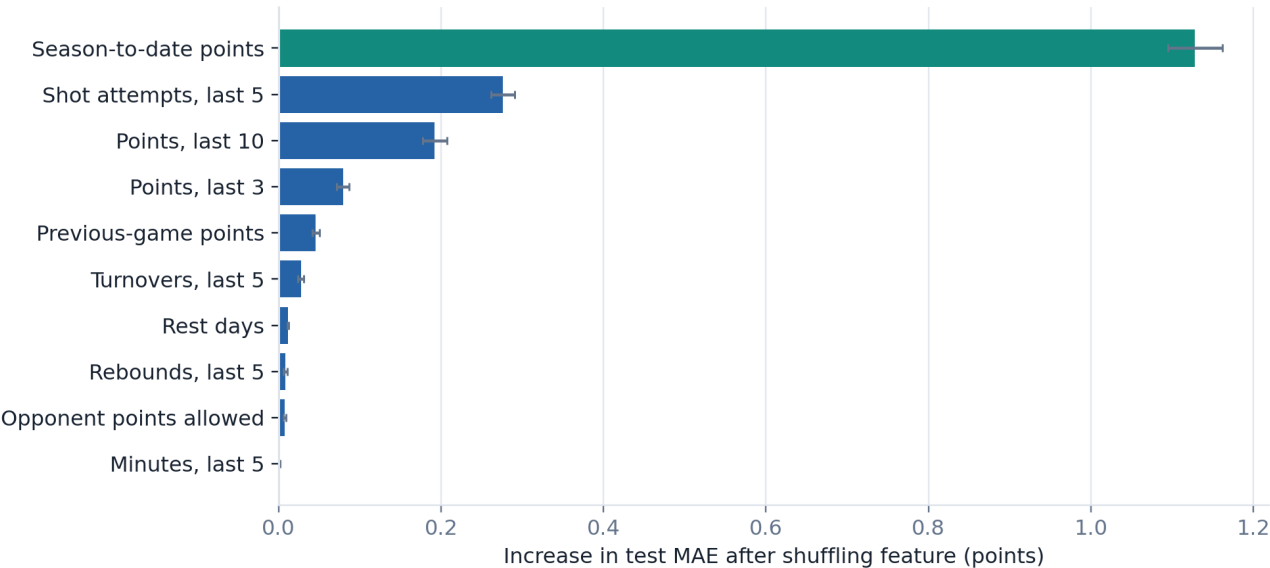
Main point: Stars miss by more points because they score more. Relative to their scoring average, their error is smaller than the error for low-volume role players.

Q1: Which inputs help the model most?

We shuffle one input at a time and measure how much the test error changes.

A player's usual scoring level is the strongest predictor

Permutation importance on held-out 2024-25 games



Source: NBA Stats API. Error bars show variation across five independent shuffles.

Figure 3. Each feature is shuffled in the held-out season. A large increase in MAE means the fitted model depended on that feature. Correlated features can share importance, so values should not be read as causal effects.

The player's season-to-date scoring level dominates. Recent shot volume and longer scoring windows add some information, while opponent points allowed, rest, home court, and back-to-backs add very little. Most of the useful information comes from the player's normal scoring level and recent role.

Q1 conclusion: A player's general scoring level is predictable, but a single game still has a lot of variation. A simple season average captures most of the information used by these models.

Limitations: the target is conditional on playing and does not model late scratches, injuries, lineup announcements, betting-market information, or expected minutes. Adding these inputs may improve the predictions.